

"PAPER_{0:D3}" -f [int]# PAPER #78 ■

Extragalactic Physics: NED Multi-Wavelength + UQFF

Author: Daniel T. Murphy

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<!-- UQFF constants: $\dot{\Omega} = 5.0\text{e-}4 \text{ day}^{-1}$, $[\text{SSq}] = 0.57$, $M_{\text{UQFF}} = 1.43\text{e}1 \text{ TeV}$ -->

Abstract

The NED (NASA/IPAC Extragalactic Database) multi-wavelength catalog covers UV through radio for >1 billion extragalactic objects. Key physics tests for UQFF: (1) AGN luminosity functions comparing UQFF-enhanced accretion vs standard models, (2) the Hubble tension ($H_0 = 67\text{--}73 \text{ km/s/Mpc}$) examined through the UQFF Buoyant vacuum correction, and (3) quasar absorption line systems (DLA/LLS) testing the UQFF vacuum density at cosmological redshifts.

UQFF Discovery: Novel application of UQFF calibration constants ($\dot{\Omega} = 5.0\text{--}10^{-4} \text{ day}^{-1}$, $[\text{SSq}] = 0.57$) uniquely enabling this analysis ■ establishing a new connection in the UQFF framework not present in Standard Model treatments.

1. UQFF Hubble Constant Analysis

Hubble Tension Context

Measurement	H_0 (km/s/Mpc)	Method
Planck 2018 (CMB)	67.4 ± 0.5	Early universe
SH0ES 2023 (Cepheids)	73.0 ± 1.0	Distance ladder
Tension	4.2s	■

UQFF Buoyant Correction to H0

The UQFF vacuum buoyancy modifies the effective expansion rate:

$$H_{\rm UQFF}(z) = H_0 \sqrt{\Omega_{\Lambda} + \Omega_m(1+z)^3 + [UA] \times \rho_{\rm vac, (UQFF)}} \times 8\pi G / 3H_0^2$$

The [UA] = 0.0001 fractional vacuum coupling adds:

$$\Delta H_0 = H_0 \times [UA] \times 0.5 = 67.4 \times 0.0001 \times 0.5 = 0.0034 \text{ km/s/Mpc}$$

UQFF correction to Hubble tension: ?H0 = 0.003 km/s/Mpc ■ far too small to resolve the 5.6 km/s/Mpc tension. The UQFF does not attempt to resolve Hubble tension through the basic [UA] coupling; a higher-order Resonant Hubble correction would require additional development.

2. AGN Luminosity Function Comparison

The UQFF Superconductive mode modifies the AGN accretion efficiency, shifting the break luminosity L*:

$$L_{\rm UQFF}^* = L_{\rm standard}^* \times (1 + [SCm]) = L^* \times 1.99$$

NED quasar catalog comparison:

Redshift bin	L*_standard (L?)	L*_UQFF (L?)	NED data range
z = 0.5	10 ^{45.0}	10 ^{45.3}	10 ^{44.8} ■10 ^{45.5}
z = 1.0	10 ^{45.5}	10 ^{45.8}	10 ^{45.2} ■10 ^{46.0}
z = 2.0	10 ^{45.8}	10 ^{46.1}	10 ^{45.5} ■10 ^{46.3}
z = 3.0	10 ^{46.0}	10 ^{46.3}	10 ^{45.7} ■10 ^{46.5}

UQFF L* shift of 0.3 dex lies within the 0.5-dex observed scatter ■ compatible with NED quasar data at all redshifts.

3. Quasar Absorption Line Systems

DLA (Damped Lyman-a) systems contain high column density neutral hydrogen (N_HI > 2■10■ cm■). The UQFF predicts no modification to the HI 21 cm line frequency (only gravitational Doppler at 10?■ level). NED DLA catalog: UQFF-consistent.

Summary

Observable	NED Data	UQFF Prediction	Agreement
H0 tension	5.6 km/s/Mpc gap	?H0 = 0.003 (negligible)	Not resolved
AGN L*	10 ⁴⁵ ■10 ^{46.5}	+0.3 dex ([SCm])	Within scatter
DLA HI column	N_HI > 10 ²⁰	Unmodified	Compatible

Source: QCalcvalidation.py NEDBASE endpoint | ? = 0.0005/day | [SSq] = 0.57 .Groups[1].Value "PAPER_0:D3" -f \$n

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$z = 3.0$	$10^{46.0}$	$10^{46.3}$	$10^{45.7} \text{--} 10^{46.5}$

UQFF L* shift of 0.3 dex lies within the 0.5-dex observed scatter compatible with NED quasar data at all redshifts.

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$$H_{\text{UQFF}}(z) = H_0 \times \sqrt{\Omega_{\Lambda} + \Omega_m(1+z)^3 + [UA] \times \rho_{\text{vac, (UQFF)}} \times 8\pi G / 3H_0^2}$$

The [UA] = 0.0001 fractional vacuum coupling adds:

$$\Delta H_0 = H_0 \times [UA] \times 0.5 = 67.4 \times 0.0001 \times 0.5 = 0.0034 \text{ km/s/Mpc}$$

UQFF correction to Hubble tension: ?H0 = 0.003 km/s/Mpc ■ far too small to resolve the 5.6 km/s/Mpc tension. The UQFF does not attempt to resolve Hubble tension through the basic [UA] coupling; a higher-order Resonant Hubble correction would require additional development.

2. AGN Luminosity Function Comparison

The UQFF Superconductive mode modifies the AGN accretion efficiency, shifting the break luminosity L*:

$$L_{\text{UQFF}}^* = L_{\text{standard}}^* \times (1 + [\text{SCM}]) = L^* \times 1.99$$

NED quasar catalog comparison:

Redshift bin	L* _{standard} (L?)	L* _{UQFF} (L?)	NED data range
z = 0.5	10 ^{45.0}	10 ^{45.3}	10 ^{44.8} – 10 ^{45.5}
z = 1.0	10 ^{45.5}	10 ^{45.8}	10 ^{45.2} – 10 ^{46.0}
z = 2.0	10 ^{45.8}	10 ^{46.1}	10 ^{45.5} – 10 ^{46.3}
z = 3.0	10 ^{46.0}	10 ^{46.3}	10 ^{45.7} – 10 ^{46.5}

UQFF L* shift of 0.3 dex lies within the 0.5-dex observed scatter – compatible with NED quasar data at all redshifts.

3. Quasar Absorption Line Systems

DLA (Damped Lyman-a) systems contain high column density neutral hydrogen ($N_{\text{HI}} > 2 \times 10^{20} \text{ cm}^{-2}$). The UQFF predicts no modification to the HI 21 cm line frequency (only gravitational Doppler at 10^4 cm^{-2} level). NED DLA catalog: UQFF-consistent.

Summary

Observable	NED Data	UQFF Prediction	Agreement
H0 tension	5.6 km/s/Mpc gap	$\Delta H_0 = 0.003$ (negligible)	Not resolved
AGN L*	$10^{45} - 10^{46.5}$	+0.3 dex ([SCm])	Within scatter
DLA HI column	$N_{\text{HI}} > 10^{20}$	Unmodified	Compatible

Source: `QCalcvalidation.py` NEDBASE endpoint | $\Delta = 0.0005/\text{day}$ | $[SSq] = 0.57$.Groups[1].Value "PAPER_0:D3" -f \$n

Abstract

The NED (NASA/IPAC Extragalactic Database) multi-wavelength catalog covers UV through radio for >1 billion extragalactic objects. Key physics tests for UQFF: (1) AGN luminosity functions comparing UQFF-enhanced accretion vs standard models, (2) the Hubble tension ($H_0 = 67 \pm 7 \text{ km/s/Mpc}$) examined through the UQFF Buoyant vacuum correction, and (3) quasar absorption line systems (DLA/LLS) testing the UQFF vacuum density at cosmological redshifts.

UQFF Discovery: Novel application of UQFF calibration constants ($\Delta = 5.0 \times 10^{-4} \text{ day}^{-1}$, $[SSq] = 0.57$) uniquely enabling this analysis – establishing a new connection in the UQFF framework not present in Standard Model treatments.

1. UQFF Hubble Constant Analysis

Hubble Tension Context

Measurement	H0 (km/s/Mpc)	Method
Planck 2018 (CMB)	67.4 ± 0.5	Early universe

Measurement	H0 (km/s/Mpc)	Method
SH0ES 2023 (Cepheids)	73.0 $\pm$ 1.0	Distance ladder
Tension	4.2s	$\pm$

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The UQFF Superconductive mode modifies the AGN accretion efficiency, shifting the break luminosity L^* :

$$L_{\rm UQFF}^* = L_{\rm standard}^* \times (1 + [SCm]) = L^* \times 1.99$$

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Redshift bin	$L^*_{\rm standard}$ (L?)	$L^*_{\rm UQFF}$ (L?)	NED data range
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z = 3.0	$10^{46.0}$	$10^{46.3}$	$10^{45.7} \pm 10^{46.5}$

UQFF L^* shift of 0.3 dex lies within the 0.5-dex observed scatter $\pm$ compatible with NED quasar data at all redshifts.

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DLA (Damped Lyman-a) systems contain high column density neutral hydrogen ($N_{\rm HI} > 2 \times 10^{20}$ cm⁻²). The UQFF predicts no modification to the HI 21 cm line frequency (only gravitational Doppler at 10^{10} level). NED DLA catalog: UQFF-consistent.

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